Do Now · retrieval

Lossy and lossless compression

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What can a firewall block?

Q2

What is non-volatile storage?

Q3

What is a bitwise mask?

Q4

Apply bitwise AND: $11011010_2 \wedge 00001111_2$. Give 8-bit answer and hexadecimal.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Lossless compression represents data more efficiently without discarding any information and reconstructs the original exactly.

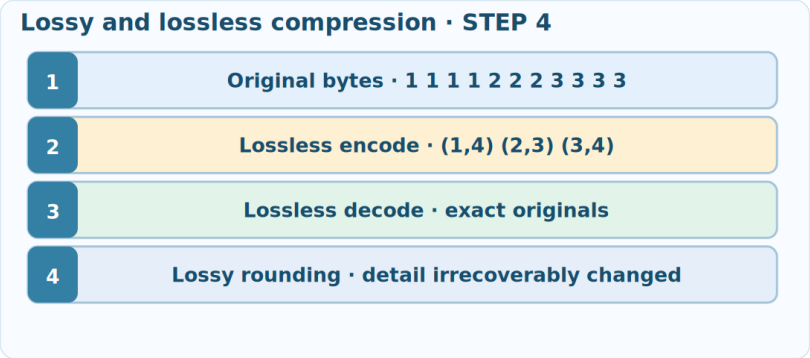
Core idea 2

Apply — Lossy methods intentionally discard information; a decoded image/audio can be perceptually similar but not bit-for-bit identical.

Core idea 3

Evaluate — Choose lossless for text, programs, records or medical numeric originals; lossy can be appropriate for delivery copies of photos/video/audio with acceptable quality reduction.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

lossless — Reconstructs original data exactly.

lossy — Permanently removes some information.

compression ratio — Original size compared with compressed size.

redundancy — Repeated/predictable information.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

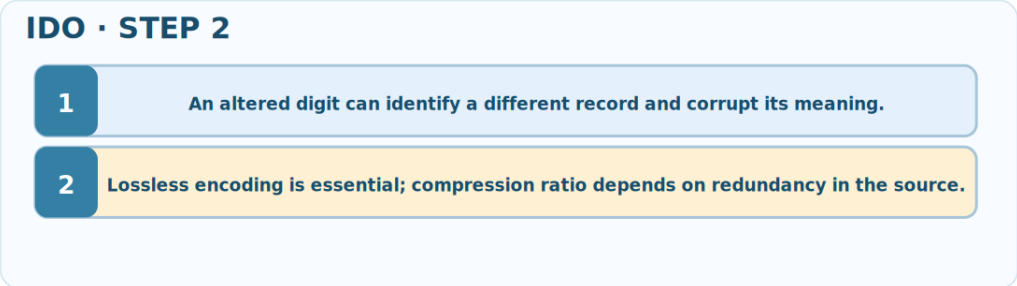
A database contains precise 7-digit identifiers. Is lossy compression suitable?

Follow the worked method

1. Predict first — An altered digit can identify a different record and corrupt its meaning.

2. Model and check — Lossless encoding is essential; compression ratio depends on redundancy in the source.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first
A lossy delivery copy may cut download size, latency and bandwidth while preserving acceptable visible quality.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define lossy and lossless compression; state one suitable use of each. [6]

Question 2

Given sequence A A A B B A, write RLE run/value pairs and show how to decode. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A school distributes music and exam marks electronically. Recommend encoding and justify trade-offs. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Lossy compression is suitable for any data if it produces the smallest file.

Use key vocabulary from memory

1. Explain lossless in your own words.

2. Explain lossy in your own words.

3. Explain compression ratio in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.